

DAA

CSA0603

DAY 3 LAB EXPERIMENTS

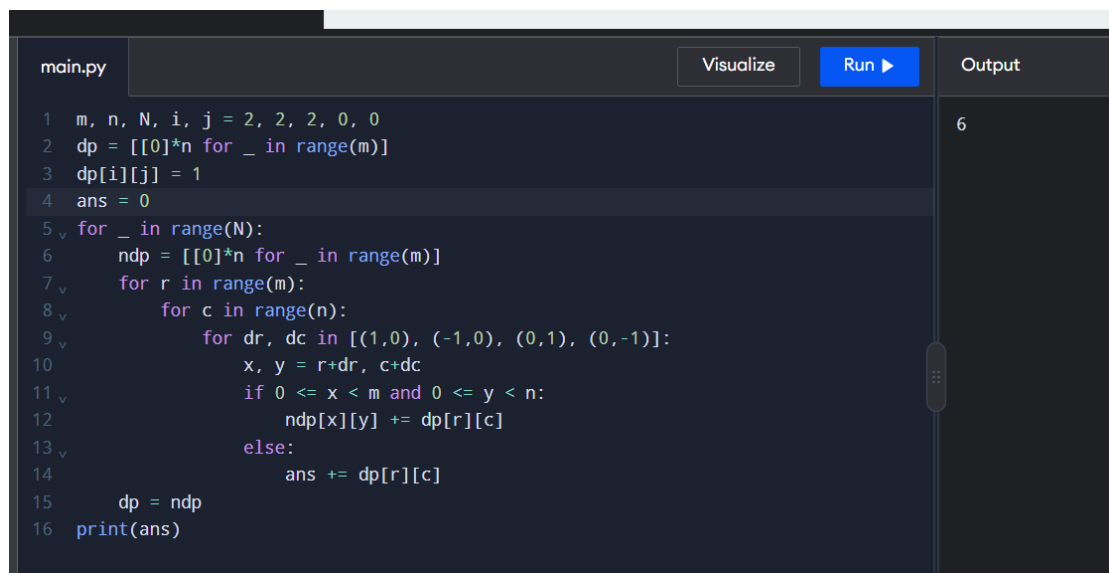
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4. Given an $m \times n$ grid and a ball at a starting cell, find the number of ways to move the ball out of the grid boundary in exactly N steps. Example:

Input: $m=2, n=2, N=2, i=0, j=0$ Output: 6

Input: $m=1, n=3, N=3, i=0, j=1$ Output: 12



```
main.py Visualize Run ▶ Output
1 m, n, N, i, j = 2, 2, 2, 0, 0
2 dp = [[0]*n for _ in range(m)]
3 dp[i][j] = 1
4 ans = 0
5 for _ in range(N):
6     ndp = [[0]*n for _ in range(m)]
7     for r in range(m):
8         for c in range(n):
9             for dr, dc in [(1,0), (-1,0), (0,1), (0,-1)]:
10                x, y = r+dr, c+dc
11                if 0 <= x < m and 0 <= y < n:
12                    ndp[x][y] += dp[r][c]
13                else:
14                    ans += dp[r][c]
15     dp = ndp
16 print(ans)
```

6

4. You are a professional robber planning to rob houses along a street. Each house has a certain amount of money stashed. All houses at this place are arranged in a circle. That means the first house is the neighbor of the last one. Meanwhile, adjacent houses have security systems connected, and it will automatically contact the police if two adjacent houses were broken into on the same night.

Examples:

Input : `nums = [2, 3, 2]`

Output : The maximum money you can rob without alerting the police is 3(robbing house 1).

(ii) Input : `nums = [1, 2, 3, 1]`

Output : The maximum money you can rob without alerting the

police is 4 (robbing house 1 and house 3).



The screenshot shows a Python Online Compiler interface. The code in the editor is as follows:

```
main.py
1 nums = list(map(int, input().split()))
2 if len(nums) == 1:
3     print(nums[0])
4 else:
5     ans = 0
6     for a in [nums[:-1], nums[1:]]:
7         p = c = 0
8         for x in a:
9             p, c = c, max(c, p+x)
10        ans = max(ans, c)
11 print(ans)
```

The output on the right is:

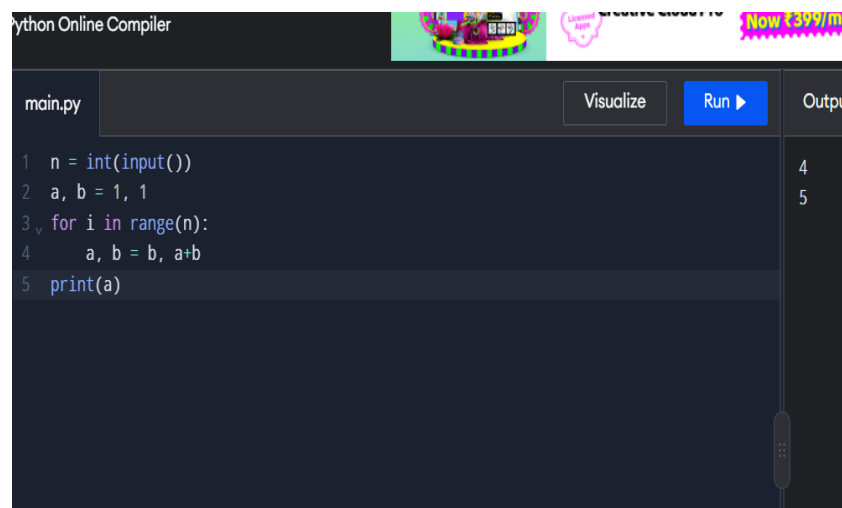
```
2 3 2
3
```

6. You are climbing a staircase. It takes n steps to reach the top. Each time you can either climb 1 or 2 steps. In how many distinct ways can you climb to the top?

Examples:

Input: $n=4$ Output: 5

Input: $n=3$ Output: 3



The screenshot shows a Python Online Compiler interface. The code in the editor is as follows:

```
main.py
1 n = int(input())
2 a, b = 1, 1
3 for i in range(n):
4     a, b = b, a+b
5 print(a)
```

The output on the right is:

```
4
5
```

7. A robot is located at the top-left corner of a $m \times n$ grid. The robot can only move either down or right at any point in time. The robot is trying to reach the bottom-right corner of the grid. How many possible unique paths are there?

Examples:

Input: m=7,n=3 Output: 28

Input: m=3,n=2 Output: 3

The screenshot shows a code editor with a file named 'main.py'. The code is a Python program that calculates the number of ways to reach the bottom-right cell of a grid of size m x n. The code is as follows:

```
1 m, n = map(int, input().split())
2 dp = [[1]*n for _ in range(m)]
3 for i in range(1, m):
4     for j in range(1, n):
5         dp[i][j] = dp[i-1][j] + dp[i][j-1]
6
7 print(dp[-1][-1])
```

There are 'Visualize' and 'Run' buttons at the top right of the editor. The 'Run' button is highlighted in blue. To the right of the code editor, there is an 'Output' panel showing the output of the program:

```
7 3
28
```